

Bernardo José Villegas Moreno

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PhD Candidate, University of Cambridge

My work advances societal adaptation to AI by measuring deployment-level externalities. I build experimental infrastructure generating evidence on harmful uses and defensive interventions: when professionals use AI assistance, how do downstream effects propagate to others? Which governance mechanisms reduce negative impacts without blocking beneficial applications? This bridges technical measurement, institutional governance, and cross-cultural validation. My Ecuador-to-Cambridge perspective enables research on how adaptation strategies must vary across contexts.

EDUCATION

University of Cambridge, Wolfson College

Cambridge, UK (2025–2030)

PhD in Human-Inspired Artificial Intelligence

TRACE Lab - Centre for Human-Inspired Artificial Intelligence (CHIA).

Supervisor: Dr. Umang Bhatt. Co-Supervisor: Prof. Anna Korhonen.

University of Edinburgh

Edinburgh, UK (2021–2023)

MSc Data Science, Technology and Innovation (Distinction)

College of Science and Engineering. Supervisor: Dr. Gian Marco Campagnolo.

Dissertation: A social data science metric for machine learning interpretability.

Pontificia Universidad Católica del Ecuador

Quito, Ecuador (2014–2019)

BA in Sociology with a minor in Social Development (First-Class Honours)

RESEARCH EXPERIENCE

University of Cambridge, Centre for Human-Inspired AI

(2025 – present)

PhD Researcher, TRACE Lab

Generating empirical evidence on AI externalities—spillover effects when one person’s AI use affects others who never consented to AI interaction. Built experimental platform enabling controlled studies of cascading social effects across approximately 5,000 participant interactions. Platform architecture supports synchronous multi-user experiments testing how AI assistance affects downstream knowledge transfer, trust calibration, and collaborative decision-making. Currently deployed by three Cambridge research teams studying interpersonal effects of AI use.

Developed theoretical taxonomy mapping first-order effects (on AI users), second-order effects (on interaction partners), and third-order effects (on institutions and markets), providing structured framework for governance assessment. Framework distinguishes mechanisms by which individual AI use generates collective consequences: through degraded knowledge transfer, shifted trust calibration, consultation displacement, and emergent institutional norms.

International Programme on AI Evaluation

(Feb – May 2025)

Student Fellow, Safety Track

Developing evaluation frameworks for second- and third-order effects of AI deployment on human collaboration—how AI mediating some interactions reshapes collaboration patterns for entire communities. Focus on cross-cultural sociotechnical safety evaluation methods.

University of Edinburgh & Alan Turing Institute

(2021–2024)

Research Collaborator

Designed interaction-based metric for ML interpretability measuring what technical xAI literature misses: whether users can calibrate trust, identify errors, and take appropriate action with AI explanations. Tracking how shared understanding emerges through conversational dynamics between developers and domain experts. Analysis reveals interpretability develops over time through sustained interaction rather than existing as inherent model property.

Pontificia Universidad Católica del Ecuador

(2019–2023)

Research Associate

Led two interdisciplinary projects combining social science and data science methods: (1) Multilevel longitudinal analysis of 2,000+ undergraduate performance trajectories identifying institutional versus individual determinants of academic outcomes (published, *Journal of Hispanic Higher Education*); (2) Network analysis of economic elite concentration across Ecuador’s top 100 corporations, mapping shareholder interlocks revealing oligarchic structures (published, *Redes*). Combined traditional

PUBLICATIONS & WORKING PAPERS

Peer-Reviewed Publications

1. Buenaño E.; Araujo L.; **Villegas B.**; Morales F. (2025) “The determinant factors of academic performance of university students in Ecuador. A multilevel longitudinal study”. *Journal of Hispanic Higher Education*. DOI: <https://doi.org/10.1177/153>
2. **Villegas, B.** & Morales F. (2021) “The economic elites of contemporary Ecuador: analysis of shareholder networks of the 100 companies with the highest income, before and at the end of Rafael Correa’s government (2007-2016)”. *Redes. Revista Hispana para el análisis de redes sociales*. 32(1): 1-15. DOI: [10.5565/rev/redes.887](https://doi.org/10.5565/rev/redes.887)
3. Jaramillo, J.; Molina, D. & **Villegas, B.** (2021) “Will Artificial Intelligence (AI) overcome professional experience? Applications of AI and Legal Analytics in International Commercial Arbitration”. In: Chipana, Catalán & Mauricio Duarte. *Arbitration and New Technologies*. Lima: Themis.

Manuscripts Under Review

1. Campagnolo G.M.; Procter R.; **Villegas B.**; van Lier F-A (2025) “Algorithmic encounters: an interactional framework for AI interpretability”. Under review.

Working Papers

1. Bhatt U.; Collins K.; **Villegas B.**; et al. (2026) “The Widening Ripples of Human-AI Interaction”. Working paper (10 pages)
2. Calcagno D.; Phua S. Z.; Devedzhiev K.; Collins K.; **Villegas B.**; et al. (2026) “Evaluating Agentic AI Decision Traces”.

CONFERENCES

- April 2026 **ACM CHI-HCXAI Workshop**, Interpretability as shared understanding: an interactional metric for evaluating ML explanations.
- May 2026 **Machine+Behavior**, The second-order effects of AI assistance in everyday knowledge transfer.
Wolfson College Cambridge Research Event, Beyond the User: How AI Assistance Reshapes Human-to-Human Interaction.

PROFESSIONAL EXPERIENCE

General Coordinator of Information

Sep 2019 – Sep 2025

Vice-Rectorate for Academic Affairs, Pontificia Universidad Católica del Ecuador

- Project manager for Ecuador’s oldest private university (30,000 students across seven campuses), leading cross-functional initiatives in academic policy, digital transformation, and institutional strategy.
- Drove digital transformation projects including AI-powered administrative tools (chatbots for student services, automated document processing), data infrastructure modernization (migrating legacy systems to cloud-based analytics platforms), and real-time dashboards for university-wide decision-making used by academic leadership.
- Led implementation of institutional analytics framework measuring student progression, faculty performance, and program effectiveness—infrastructure enabling evidence-based policy decisions on resource allocation, curriculum design, and student support interventions.
- Managed stakeholder coordination across seven autonomous campus administrations, reconciling competing priorities and building consensus for university-wide digital initiatives. Developed institutional data governance protocols balancing transparency with privacy protection.
- Progressive advancement through four roles demonstrating increasing scope: Data Analyst → Scholarships Specialist → Student Affairs Coordinator → General Coordinator. Trajectory required navigating complex organizational politics, building coalitions across administrative silos, and translating technical capabilities into institutional value.

Associate Data Intelligence Consultant

Feb 2019 – Aug 2019

Diozen Integral Consulting by Results, Quito, Ecuador

- Delivered data analysis and visualization solutions for clients across insurance, healthcare, and social entrepreneurship sectors, translating business requirements into technical implementations.
- Key projects: Business intelligence dashboards for Tecniseguros (insurance sector) tracking claims processing efficiency and risk assessment; operational analytics for healthcare providers (InstaSalud, PUCESalud) measuring patient flow and resource utilization; impact measurement frameworks for U.S. Embassy-funded social ventures evaluating program effectiveness.
- Developed client-facing reporting systems using Python, R, and Tableau, enabling non-technical stakeholders to access real-time performance metrics and make data-informed operational decisions.

TECHNICAL & METHODOLOGICAL SKILLS

Computational	Python, R, JavaScript, Empirica experimental platform development, NLP (transformers, topic modeling), network analysis, statistical modeling (multilevel models, longitudinal analysis), web-based experiment design, data visualization (Tableau, D3.js)
Social Science	Ethnographic methods, qualitative coding, survey design, interview protocols, participatory observation, institutional analysis
Sociotechnical Integration	Experimental design for human-AI interaction studies, mixed-methods evaluation frameworks, cross-cultural research adaptation, empirical AI governance research, sociotechnical system evaluation
Languages	Spanish (Native), English (C1 IELTS — Proficient)

TEACHING EXPERIENCE

King’s College, Cambridge	Concurrent & Distributed Systems (Michaelmas 2025)
	Computer Networking (Lent 2026)

FELLOWSHIPS & AWARDS

2026	Student Fellow , International Programme on AI Evaluation: Capabilities and Safety
2026–2028	Student Fellow , .406 Ventures
2023	MSc Distinction, University of Edinburgh
2021	Fondo Público Award, Best dissertation published (PUCE)
2014	SNNA Scholarship, Awarded for scoring in top 1% nationally on ENES exam

LEADERSHIP & SERVICE

2018–2020	General Secretary , Federation of Private University Students of Ecuador (FEUPE)
2018–2019	President , Student Federation, Pontificia Universidad Católica del Ecuador (FEUCE)